Name:_____ Class:_____ Student number:_____

Why do young shootsgrow towards light?

Clues from everyday life

Why do coleoptiles（胚芽鞘） near a window bend towards light?

This experiment uses oat coleoptiles（胚芽鞘）. A coleoptile（胚芽鞘） is a sheath that covers and protects the young leaves inside. Its fairly straight shape makes changes easy to observe.

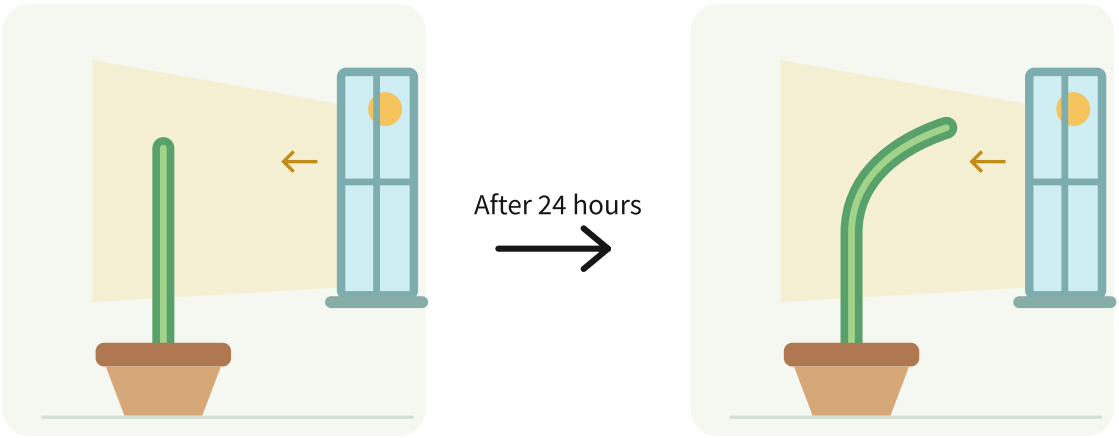


Illustration of growth changes; 24 hours is the time chosen for this situation.

The school gardening club notices that coleoptiles（胚芽鞘） near a window gradually bend towards it. They know that plants grow towards light, but they do not know which part senses light or what makes the plant bend. Can you find clues through experiments?

Your first observations

Describe the changes you can observe. How do the coleoptiles（胚芽鞘）' shape and growth direction change?

(Record observations only. You do not need to explain them here.)

Research question

Which part of a coleoptile（胚芽鞘） senses light from one side and affects its growth direction?

01 • Build a hypothesis

Complete your hypothesis

If the coleoptile（胚芽鞘）'s （Tip ／ Part below the tip） is covered to block light while the other parts still receive light, it will （still bend towards light ／ show no clear bending towards light）.

My reason (required; there is no single answer)

Your full original hypothesis and reason will be saved before your first experiment as part of your inquiry record.

02 • Plan a fair test

Change only the light-blocking treatment. Compare no light blocked, light blocked at the tip and light blocked below the tip. All three setups keep their tips, and other conditions stay the same.

Independent variable(the factor changed on purpose)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Dependent variable(the result measured or observed)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Controlled variables(the factors kept the same)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Revision (if needed)

03 • Experiment assumptions

What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

☐ The coleoptiles（胚芽鞘） start in similar growth conditions.

☐ The light-blocking cap and sleeve do not limit elongation（延長） or bending.

☐ The treatments do not cause clear temperature differences between setups.

☐ If the plants are the same type, their starting size does not need to be controlled.

04 • The control group

To find out whether blocking light affects the coleoptile（胚芽鞘）'s growth direction, how would you design the control group?

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

05 • Experiment setup

Draw and label your experiment setup

Use your experiment variables to draw and label your setup.

(Hint: How many setups do you need? Which are the test groups and the control group?)

Oat coleoptiles（胚芽鞘） × 3; Light source on one side × 3; Opaque（不透光） cap × 1; Light-blocking sleeve for the lower part × 1; Ruler × 1; Timer × 1

Draw, upload a photo or write your plan, then press Save below to save them together.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

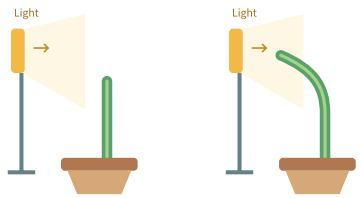
Written plan (may replace a drawing): describe setups A, B and C, how light is blocked and the conditions kept the same.

Light from the same side, three light-blocking treatments

The light is on the left for every setup. Before treatment, the coleoptiles（胚芽鞘） are similar in type, starting height and growth stage. Light intensity, exposure time, temperature and water supply are the same. The opaque（不透光） sleeve covers the whole part below the tip, leaving the tip exposed. The sleeve can bend as the plant grows and does not limit elongation（延長）.

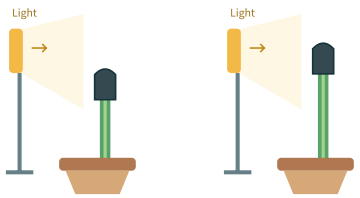
The animation speeds up growth and shows a typical response. The time and data are teaching simulations, not real measurements. The model does not show differences between individual plants.

A • No light blocked



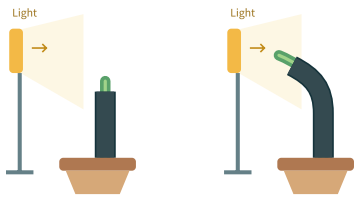
Before growth　After growth

B • Light blocked at the tip



Before growth　After growth

C • Light blocked below the tip



Before growth　After growth

My observation record

Setup	Treatment	First record • Bending direction	Revision (if needed)
A	No light blocked	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）
B	Light blocked at the tip	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）
C	Light blocked below the tip	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）

Your observations are saved. If you repeat the experiment and revise them, your first observations and action record are also kept.

Use the main comparisons to draw conclusions

1. Compare A and B: after **blocking light at the tip**, what changes?（No clear bending towards the light. ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend?（It can still bend towards light. ／ No clear bending towards the light.）

3. Which part may **sense light from one side**?（Tip ／ Only the lower part）

04 • Extension 1

Does the tip also affect coleoptile（胚芽鞘） growth / elongation（延長）, as well as sensing light?

The light-blocking comparisons helped us confirm the part of the plant that senses light. Could the tip have another role? Add a new setup and change only whether the tip is present: compare intact A with D, whose tip has been removed.

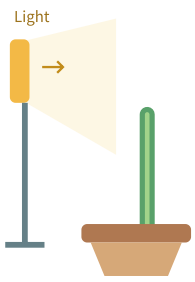
After removing the tip, what elongation（延長） do you predict compared with a complete coleoptile（胚芽鞘）?（Will not elongate（延長） ／ Similar elongation（延長） ／ More elongation（延長））

My reason

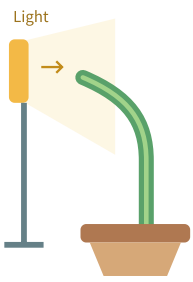
Your original extension 1 prediction and reason are saved before the first experiment. The animation and 24-hour period are teaching simulations.

A and D: compare before and after growth

A • No light blocked

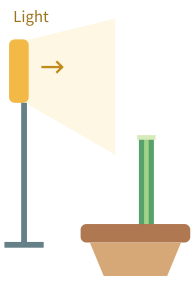


Before growth

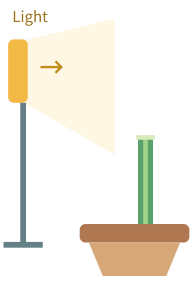


After growth

D • Tip removed



Before growth



After growth

Setup	Treatment	First record • Elongation（延長）	Revision (if needed)
A	No light blocked	（Yes ／ No）	（Yes ／ No）
D	Tip removed	（Yes ／ No）	（Yes ／ No）

Experimental conclusion

1. What does comparing elongation（延長） in A and D support?（The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）. ／ The presence of the tip does not affect the coleoptile（胚芽鞘）’s growth / elongation（延長）.）

The tip may produce and **pass down substance X**, promoting **elongation（延長）** below; this idea still needs to be tested.

04 • Extension 2

Can substance X transfer a growth-promoting effect? How does its position affect bending?

Researchers have prepared **agar（瓊脂） that has been in contact with coleoptile（胚芽鞘） tips** to test whether this effect can be **transferred** and how its position **affects bending**.

Extension 2 materials: oat coleoptiles（胚芽鞘） with tips removed × 4, growing containers × 4, blank agar（瓊脂） × 1, treated agar（瓊脂） × 3, timer × 1.

Both types of agar（瓊脂） look the same. **Blank agar（瓊脂） has not been in contact with** coleoptile（胚芽鞘） tips; **Treated agar（瓊脂） has been in contact with** coleoptile（胚芽鞘） tips. Some substances can enter and pass through agar（瓊脂）.

If **treated agar（瓊脂）** is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, how do you predict it will grow?（Bends left ／ Bends right ／ Stays fairly straight）

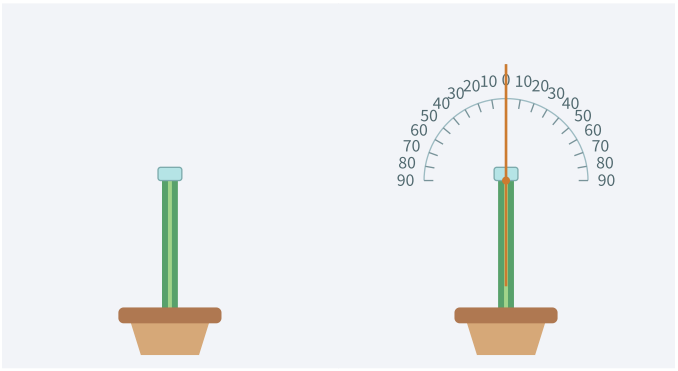
My reason

Four extension 2 setups in darkness

All coleoptiles（胚芽鞘） have their tips removed, start at a similar size and are kept in darkness. Agar（瓊脂） size, growing time, temperature and water supply are the same. You can view them before and after growth. Placing agar（瓊脂） on a coleoptile（胚芽鞘） does not physically hold it down.

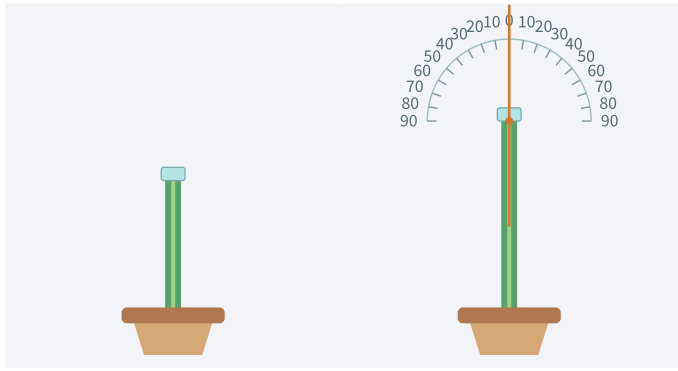
Growing time, gradual growth animations and angles are teaching simulations, not real measurements or exact experimental constants.

E • Blank agar（瓊脂） • Centre



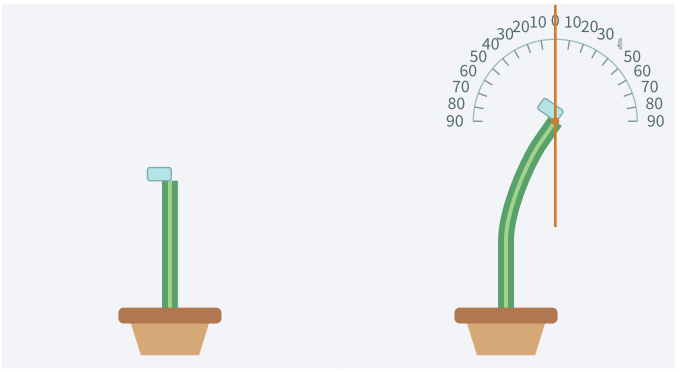
Darkness • Before growth　　Darkness • After growth

F • Treated agar（瓊脂） • Centre



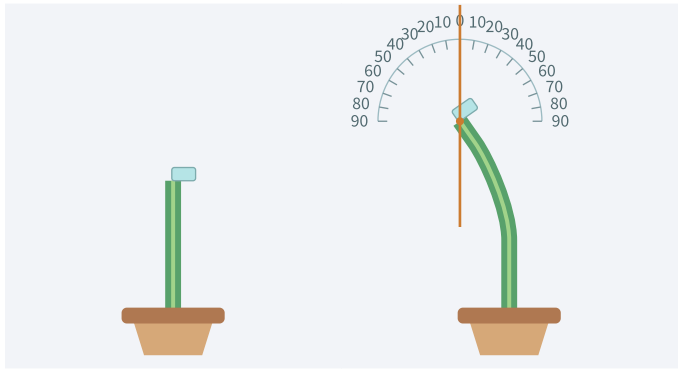
Darkness • Before growth　　Darkness • After growth

G • Treated agar（瓊脂） • Left



Darkness • Before growth　　Darkness • After growth

H • Treated agar（瓊脂） • Right



Darkness • Before growth　　Darkness • After growth

Setup	Treatment	Angle (°)	First record • Bending direction	Revision (if needed)
E	Blank agar（瓊脂） • Centre	_____ °	（Bends left ／ Bends right ／ No clear bending）	_____ °（Bends left ／ Bends right ／ No clear bending）
F	Treated agar（瓊脂） • Centre	_____ °	（Bends left ／ Bends right ／ No clear bending）	_____ °（Bends left ／ Bends right ／ No clear bending）
G	Treated agar（瓊脂） • Left	_____ °	（Bends left ／ Bends right ／ No clear bending）	_____ °（Bends left ／ Bends right ／ No clear bending）
H	Treated agar（瓊脂） • Right	_____ °	（Bends left ／ Bends right ／ No clear bending）	_____ °（Bends left ／ Bends right ／ No clear bending）

Analyse clues about substance X

1. What does the comparison of E and F support?（The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support?

◯ The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

◯ The position of treated agar（瓊脂） **does not** affect the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

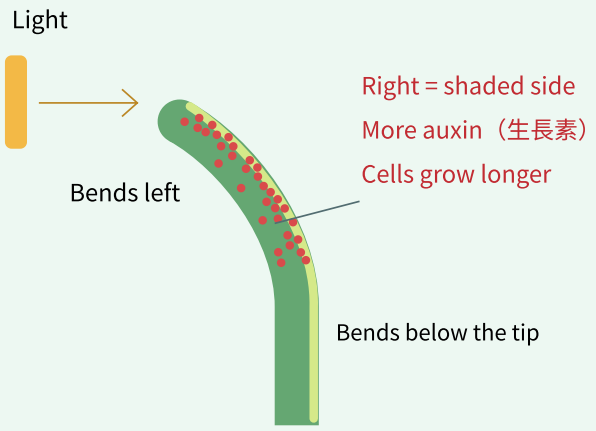
Name:_____ Class:_____ Student number:_____

From evidence to explanation

How plants grow towards light

1. Directional plant growth in response to light direction is called **"phototropism（向光性）"**; growth towards light is called "positive phototropism（向光性）".
2. In this oat coleoptile（胚芽鞘） model, **Tip** helps **sense light from one side**.
3. Other research helps us understand the substance called "substance X" in this inquiry as **Auxin（生長素）**. Auxin（生長素） is a hormone that affects plant growth; the tip can produce a growth-promoting effect that **passes downwards**.
4. With light from one side, **more auxin（生長素） is found on the shaded side, so cells on that side elongate（延長） more, making the coleoptile（胚芽鞘） bend towards light**.

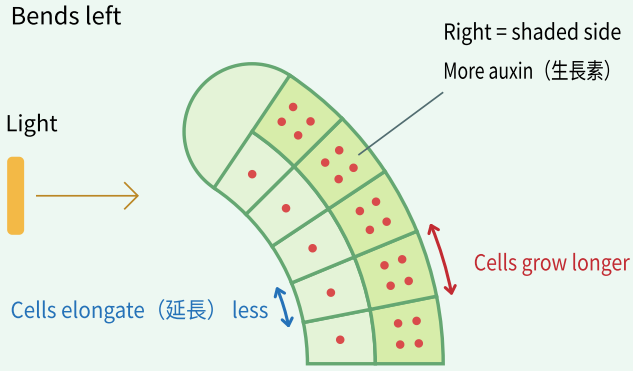
1 • Light from the left



Tip senses light → growth signal passes down

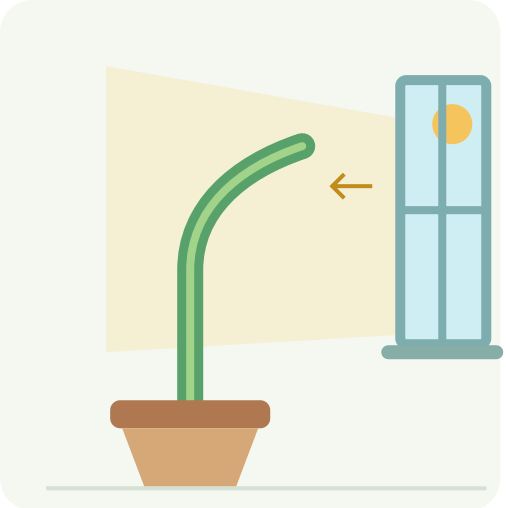
- Red dots = auxin（生長素）

2 • Light from the left: enlarged cells



More elongation（延長） on the shaded side → bends left

Learning reflection



After 24 hours

Use what you have learnt from the learning points to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.

Teacher guide and distribution order

1. Give A first. Preserve the original hypothesis, reason and design before giving B.
2. Complete B in order: main inquiry, tip comparison, then agar（瓊脂） inquiry. Preserve each prediction and reason before showing its results. Cover results not yet reached.
3. Fix the answers to all three inquiries before giving C. Keep D and the complete teacher pack for teachers only.
4. Keep original answers and record revisions separately. Do not convert time, clicks or completion into ability scores.
5. Static diagrams use the VL teaching model, not real experimental data. Students interpret directions and measure angles themselves. Do not add calculations for quantities not measured in the VL.

Research procedure alignment

Under proposal 4.1 / 4.3.2, both groups use identical scenarios, concepts and questions before formal teaching. Keep four-week activity worksheets separate from the 10-item SPSAI and 4-item CKT pre/post-tests. Do not add experimental-group opinion questions or interviews.

Paper learners read static data and use a protractor; VL learners use animation and an adjustable pointer. These operations are not identical. The researcher must check equal time and data access, original-answer copies, scoring consistency and comparability of skill evidence.

Teacher check: evidence limits

The tip and agar（瓊脂） models do not identify substance X on their own, nor directly measure auxin（生長素） distribution under light from one side. These limits belong to the teacher guide; removed student questions are not restored.

Independent variable	Light-blocking treatment
Dependent variable	The coleoptile（胚芽鞘）’s bending response
Controlled variables	Light direction and intensity; Coleoptile（胚芽鞘） type, starting height and growth stage; Time exposed to light; Temperature and water supply

Experiment assumptions

- The coleoptiles（胚芽鞘） start in similar growth conditions.
- The light-blocking cap and sleeve do not limit elongation（延長） or bending.
- The treatments do not cause clear temperature differences between setups.

A · No light blocked	Bends left
B · Light blocked at the tip	No clear bending
C · Light blocked below the tip	Bends left
A · No light blocked · Elongation（延長）	Yes
D · Tip removed · Elongation（延長）	No
E · Blank agar（瓊脂） · Centre	0°（Teaching model）; No clear bending
F · Treated agar（瓊脂） · Centre	0°（Teaching model）; No clear bending
G · Treated agar（瓊脂） · Left	35°（Teaching model; ±2°）; Bends right
H · Treated agar（瓊脂） · Right	35°（Teaching model; ±2°）; Bends left

3. Which part may sense light from one side?	Tip
1. Compare A and B: What changes when light is blocked at the tip?	No clear bending towards the light.
2. Compare A and C: Does it still bend when the lower part receives no direct light?	It can still bend towards light.
1. What does comparing elongation（延長）in A and D support?	The presence of the tip also affects the coleoptile（胚芽鞘）' s growth / elongation（延長）.
1. What does the comparison of E and F support?	The tip may produce a growth-promoting effect that can be transferred.
2. What does the comparison of F, G and H support?	The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

Learning reflection

The light source is on the right, and the coleoptile（胚芽鞘） tip senses light from one side. Auxin（生長素） moves towards the left (shaded side) and passes downwards, causing cells on the left to elongate（延長） more. The coleoptile（胚芽鞘） therefore bends towards the light on the right, showing positive phototropism（向光性）. The teacher reviews this; it is not automatically marked wrong.

Assessment item	Maximum score	Criteria / notes
Cross-VL calibration status		This draft rubric has not been validated or calibrated across VLs. Do not convert time, completion rate or clicks directly into ability scores.
First observations	2	2: accurately describes the coleoptile（胚芽鞘）, light from one side and bending; 1: partial description; 0: no relevant description.
Qualitative observations	2	The bending directions of A, B and C in the main inquiry total 1 mark; the elongation（延長） observations of A and D in extension 1 total 1 mark. Score older records using their original questions and models. Preserve original observations.
Variable classification	4	Independent and dependent variables: 1 each. Controlled variables: 2 for all correct choices with none extra; 0 for a wrong or missing choice.
Original hypothesis and reasons	2	2: testable predictions and reasonable reasons for all three stages; 1: partly complete; 0: no testable content. Do not deduct marks just because a prediction differs from the model. Score older records using their original inquiry.
Experiment assumptions	1	1: select similar initial conditions, no mechanical restriction and no clear temperature difference, without selecting that the same species needs no size control. Use the original options for older records.
Control group	1	1: compare uncovered A with B covered at the tip and C covered below the tip, keeping other conditions the same. Review the intact A versus cut-tip D comparison in extension 1 as additional evidence. Use the original design for older records.
Experimental operation	2	Extension 2: accept only 0° for E and F and 33–37° (35° ±2°) for G and H; 0.5 per setup. The teacher still assesses the measurement method and setup quality. Do not score clicks or time.
Experiment setup	2	2: clearly shows the light-blocking treatments for A, B and C, light from one side and controlled conditions; 1: partly complete; 0: no valid comparison. Use the original design for older records.
Comparison conclusions	2	The 3 main questions, 1 extension 1 question and 3 extension 2 questions total 2 marks, proportional to correct answers. Each blank in extension 2 question 3 is one third of that question; its total weight equals any other comparison question. Use original questions for older records.
Data interpretation	2	Assess the learning reflection: 2 for linking light on the right, sensing at the tip, more auxin（生長素） on the left (shaded side) passing down, more cell elongation（延長） on the left and bending right; 1 for partial links; 0 for no relevant explanation. The teacher assesses this, without automatic keyword scoring. Use original questions for older records.
Reflection expression	2	2: a clear, coherent cause-and-effect explanation of the coleoptile（胚芽鞘） bending right in the context diagram; 1: partly clear; 0: no relevant reflection. For older records, assess original ideas, comparison evidence and revisions.
Data recording and evidence communication	2	The teacher assesses clear communication of angles, directions and comparison evidence: 2 for clear and complete; 1 for partly clear; 0 for no relevant communication. Retain original chart scoring for older records.

Assessment item	Maximum score	Criteria / notes
Terminology and light-sensing part	2	Teacher assessment: 1 each for correct use of phototropism（向光性） / positive phototropism（向光性） and sensing light at the tip, considering inquiry answers and reflection.
Light-sensing and growing parts	2	2: correctly links unequal elongation（延長） on the two sides with bending direction, including bending left with light on the left and bending right with agar（瓊脂） on the left; 1: partly correct; 0: no demonstrated understanding.
Auxin（生長素） mechanism	2	2: explains that auxin（生長素） is a plant hormone and the tip produces a growth-promoting effect that can pass downwards; 1: partly correct; 0: no demonstrated understanding.
Context application / earlier evidence revision	2	2: correctly applies learning points to light on the right, explaining that the left is shaded and elongates（延長） more, causing bending right; 1: partly correct; 0: no relevant application. Use original questions for older records.
Total score	32	Six SPS categories, 4 each, total 24; new knowledge totals 8. Show the overall score only after all 11 teacher scores and reflection are complete (10 teacher scores for older records).
Blank versus zero		A blank means awaiting assessment; 0 is an explicit zero score. Excel recalculates formulas when opened.
Saving teacher scores		Teacher scores exist only in the saved Excel file. Downloading again does not import teacher scores from an earlier file.